

table_1 Comparision of models on birds sky video:

Shows how as model increases results of detection(mAP) and tracking(IDF1, MOTA, MOTP, num_switches) improve and inference time on cpu increases. used DeepSORT.

(+ decision of using yolov11x.pt to make ground truth labels for our data).

mAP50	mAP50-95	מודל
0.068	0.0306	bestbirdsonly.pt
0.0274	0.0173	yolov11n.pt
0.0537	0.0236	yolov11m.pt
0.0476	0.022	yolov11x.pt

Inference Results (Inference Results DeepSORT on birds_sky.mp4 video)							
model	idf1	mota	motp	mAP50-95	mAP50	num_switches	Inference_time (on CPU)
Yolov11n_trained	0.61	0.76	0.39	0.181	0.607	38	2 minute, 31 sec
yolov11n	0.70	0.88	0.12	0.692	0.904	36	2 minutes, 48 sec
yolov11s	0.82	0.93	0.09	0.769	0.95	23	4 minutes, 19 sec
yolov11m	0.92	0.98	0.05	0.916	0.977	5	8 minutes, 32 sec
yolov11l	0.98	0.99	0.04	0.945	0.987	2	10 minutes, 19 sec
Yolo11n best trained	0.98	0.98	0.13	0.73	0.97	1	2 minutes, 58 sec
Yolov11x (used as ground truth)	1.0	1.0	0.0	1.0	1.0	0	19 minutes, 31 sec
bestbirdsonly.pt	0.7875	0.947	0.150	0.675	0.947	21	2 minutes, 56 sec
hyperparam_n.pt	0.554	0.801	0.618	0.267	0.689	62	2 minutes, 58 sec
bestbirdsonly.pt + best parameters	0.98	0.98	0.13	0.73	0.97	1	2 min, 58 sec

table_2 Optimized parameters tracking vs. default:

Shows better performance after optimization of parameters (with random permutations and grid search using sklearn functions):

Inference Results DeepSORT on birds_sky.mp4 video							
model	idf1	mota	motp	mAP50-95	mAP50	num_switches	Inference_time (on GPU)
bestbirdsonly.pt	0.7875	0.947	0.150	0.675	0.947	21	2 minutes, 56 sec
hyperparam_n.pt	0.554	0.801	0.618	0.267	0.689	62	2 minutes, 58 sec
bestbirdsonly.pt + best parameters	0.98	0.98	0.13	0.73	0.97	1	2 min 58 sec

table_3 Tracking algorithms performance comparision:

SORT, BotSORT, ByteTrack, DeepSORT, OcSORT each one with its best parameters.

shows OcSORT and DeepSORT are generally best but Ocsort has better inference time than DeepSORT across videos.

Inference Results SORT (model: bestbirdsonly)							
video	idf1	mota	motp	mAP50-95	mAP50	num_switches	Inference_time (on CPU)
Birds flying in the sky preview	0.8	0.94	0.27	0.391	0.888	18	1 min, 16 sec
Birds Beach	0.956	0.978	0.005	NA	NA	12	1 minute, 4 sec
Similar Birds	0.97	0.96	0.084	0.826	0.969	15	1 minute, 27 sec
Oclusion birds	0.84	0.885	0.18	0.6	0.914	4	1 minute, 13 sec
Desert birds	0.765	0.919	0.153	0.641	0.928	11	2 min, 4 sec

Inference Results ByteTrack (model: bestbirdsonly)							
video	idf1	mota	motp	mAP50-95	mAP50	num_switches	Inference_time (on CPU)
Birds Sky	0.97	0.94	0.14	0.681	0.951	8	1 min, 26 sec
Birds Beach	0.59	0.79	0.27	0.404	0.832	35	51 sec
Similar Birds	0.9	0.964	0.108	0.745	0.979	12	1 min, 22 sec
Oclusion birds	0.88	0.87	0.22	0.475	0.912	4	1 min, 13 sec
Desert birds	0.866	0.927	0.03	0.918	0.951	8	2 min, 4 sec

Inference Results DeepSORT (model: bestbirdsonly)							
video	idf1	mota	motp	mAP50-95	mAP50	num_swit ches	Inference_time (on CPU)
Birds Sky	0.98	0.98	0.13	0.73	0.97	1	2 min 58 sec
Birds Beach	0.24	0.586	0.289	0.345	0.782	126	1 min, 14 sec
Similar Birds	0.899	0.965	0.137	0.745	0.979	9	2 min, 35 sec
Oclusion birds	0.93	0.892	0.22	0.489	0.912	2	1 min, 59 sec
Desert birds	0.979	0.958	0.01	0.974	0.974	0	2 min 38 sec

Inference Results BotSORT (model: bestbirdsonly)							
video	idf1	mota	motp	mAP50-95	mAP50	num_switches	Inference_time (on CPU)
Birds flying in the sky preview	0.92	0.96	0.27	0.381	0.884	3	1 minute, 30 sec
Birds beach	0.67	0.8	0.173	0.617	0.898	34	1 min 0 sec
Similar Birds	0.87	0.97	0.001	NA	NA	16	1 min, 48 sec
Oclusion birds	0.797	0.87	0.17	0.625	0.923	3	1 minute, 35 sec
desert_birds	0.9	0.942	0.139	0.662	0.94	4	3 min, 30 sec

Inference Results OcSORT (model: bestbirdsonly)							
video	idf1	mota	motp	mAP50-95	mAP50	num_switches	Inference_time (on CPU)
Birds_Sky 17 sec	0.95	0.95	0.27	0.36	0.85	5	1 minute and 23 sec
Oclusion birds 15 sec	0.93	0.89	0.16	0.57	0.89	2	1 minute and 17 sec
Birds_beach 10 sec	0.7814	0.88	0.244	0.455	0.897	4	52 sec
Similar_birds 10 sec	0.943	0.95	0.12	0.894	0.97	6	1 min 33 sec
desert_birds	0.91	0.90	0.121	0.613	0.914	9	4 min 35 sec

table_4 Final inference - OcSORT(with best parameters) inference on all data (5 videos, 3 models):

Shows our model is better than [yolov11n.pt](#) and [yolov11m.pt](#) across all videos and metrics!

Inference Results yolo11n.pt					
video	idf1	mota	motp	num_switches	Inference_time (on CPU)
Birds Sky	0.86	0.87	0.28	23	76.18 sec
Birds beach	0.80	0.74	0.73	31	40 sec
Similar birds	0.48	0.79	0.47	35	77.36 sec
Occlusion birds	0.59	0.76	0.20	19	69.72 sec
Desert birds	0.86	0.87	0.35	5	255 sec

Inference Results yolo11m.pt					
video	idf1	mota	motp	num_switches	Inference_time (on CPU)
Birds Sky	0.98	0.98	0.24	0	382 sec
Birds beach	0.90	0.77	0.78	34	205 sec
Similar birds	0.70	0.899	0.16	29	282 sec
Occlusion birds	0.66	0.76	0.22	27	366 sec
Desert birds	0.88	0.91	0.2	8	785 sec

Inference Results bestbirdsonly.pt					
video	idf1	mota	motp	num_switches	Inference_time (on CPU)
Birds_Sky	0.95	0.96	0.27	5	71 sec
Birds beach	0.82	0.88	0.24	5	42 sec
Similar birds	0.94	0.95	0.12	6	74 sec
Occlusion birds	0.93	0.89	0.16	2	65 sec
Desert birds	0.88	0.92	0.157	8	261 sec